

DONKIN, Prof. On a Class of Differential Equations, including those which occur in Dynamical Problems. Part II.—*Phil. Trans.* 1855, pp. 299–358. Received Feb. 17th, read March 22nd, 1855. 1855.

LIOUVILLE. Rapport sur un mémoire de M. Bour concernant l'intégration des équations différentielles de la Mécanique Analytique.—*Comptes Rendus*, t. XL. p. 661, séance du 26 Mars, 1855;—*Liouville*, t. XX. pp. 135–136. 1855.

BOUR. Sur l'intégration des équations différentielles de la Mécanique Analytique (extrait d'un mémoire présenté à l'Académie le 5 Mars, 1855).—*Liouville*, t. XX. pp. 185–200. 1855. [And *Mém. Savants Étrang.* t. XIV. pp. 35–58, 1856.]

LIOUVILLE. Note à l'occasion du mémoire précédent de M. Edmond Bour.—*Liouville*, t. XX. pp. 201–202. 1855.

BRIOSCHI. Sopra una nuova proprietà degli integrali di un problema di dinamica.—*Tortolini*, t. VI. pp. 430–432. 1855.

BERTRAND. Mémoire sur quelque'une des formes les plus simples que puissent prendre les intégrales des équations différentielles du mouvement d'un point matériel.—*Liouville*, t. II. (2^e série), pp. 113–140. 1857.

[Marginal note in scan: "See [298] of these papers (Vol. 4)."]

196.

NOTE SUR UN PROBLÈME D'ANALYSE INDÉTERMINÉE.

[From the *Nouvelles Annales de Mathématiques* (Terquem and Gerono), t. XVI. (1857), pp. 161–165.]

EULER a donné dans le Mémoire: *Regula facilis problemata Diophantea per numeros integros expedite solvendi* (*Comment. Arith. Coll.*, t. II., p. 263) la solution que voici de l'équation indéterminée

$$\alpha x^2 + \beta x + \gamma = \zeta y^2 + \eta y + \theta; \quad (1)$$

en supposant que l'on ait la solution $x = a$, $y = b$ de manière que

$$\alpha a^2 + \beta a + \gamma = \zeta b^2 + \eta b + \theta,$$

et en posant

$$s = \sqrt{\alpha \zeta r^2 + 1},$$

où r est une quantité quelconque, l'équation sera satisfaite par les valeurs

$$\begin{aligned} x &= sa + \zeta rb + \frac{(s-1)\beta}{2\alpha} + \frac{1}{2}r\eta, \\ y &= \alpha ra + sb + \frac{(s-1)\eta}{2\zeta} + \frac{1}{2}r\beta; \end{aligned}$$

en effet, on voit sans peine que ces valeurs donnent identiquement

$$\alpha x^2 + \beta x + \gamma - (\zeta y^2 + \eta y + \theta) = \alpha a^2 + \beta a + \gamma - (\zeta b^2 + \eta b + \theta) = 0. \quad (2)$$

En supposant de plus que les coefficients α , β , γ , ζ , η , θ soient des nombres entiers tels que $\alpha\zeta$ soit un entier positif non carré, on peut toujours déterminer le nombre entier r de manière que s soit un nombre entier; cela étant, et en supposant que a , b soient des entiers, il est évident que x , y seront des nombres rationnels. Euler a de plus remarqué que l'on peut toujours faire en sorte que x , y soient des nombres entiers. En effet, si les formules donnent $x = a'$, $y = b'$ des valeurs non entières, en

substituant dans les formules au lieu de a, b les valeurs a', b' , on obtiendra pour x, y des valeurs entières; cela se vérifie sans peine.

L'équation indéterminée (2) rentre dans celle-ci

$$(a, b, c, f, g, h \mid x', y', z')^2 = (a, b, c, f, g, h \mid x, y, z)^2; \quad (3)$$

en supposant que la forme ternaire

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \xi, \eta, \zeta)^2$$

se transforme en elle-même au moyen d'une substitution linéaire quelconque, on peut supposer que cette substitution soit telle que l'on ait $z' = z$; cela étant, en écrivant $z' = z = 1$ et en mettant de plus $h = 0$, l'équation (3) se réduit évidemment à une forme telle que l'équation (2). Or on peut trouver par la méthode générale de M. Hermite la solution convenable de l'équation (3). En supposant, comme à l'ordinaire,

$$\mathfrak{A} = bc - f^2, \dots, \quad \mathfrak{F} = gh - af, \dots,$$

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

il faut pour cela écrire

$$x' = 2\xi - x, \quad y' = 2\eta - y, \quad z' = 2\zeta - z,$$

et

$$\begin{aligned} ax + hy + gz &= a\xi + h\eta + g\zeta && - q\mathfrak{G}\eta + q\mathfrak{H}\zeta, \\ hx + by + fz &= h\xi + b\eta + f\zeta + q\mathfrak{F}\xi && - q\mathfrak{H}\zeta, \\ gx + fy + cz &= g\xi + f\eta + c\zeta - q\mathfrak{F}\xi + q\mathfrak{G}\eta, \end{aligned}$$

où q est une quantité arbitraire. En effet, en multipliant ces équations par ξ, η, ζ et en ajoutant, on obtient

$$(a, b, c, f, g, h \mid \xi, \eta, \zeta)(x, y, z) = (a, b, c, f, g, h \mid \xi, \eta, \zeta)^2,$$

et, au moyen de cette équation et des valeurs

$$x' = 2\xi - x, \quad y' = 2\eta - y, \quad z' = 2\zeta - z,$$

on forme tout de suite l'équation (3). De plus, en multipliant les trois équations par $\mathfrak{G}, \mathfrak{F}, \mathfrak{C}$ et en ajoutant, on obtient $Kz = K\zeta$, c'est-à-dire $z = \zeta$ et de là $z' = z$.

Cela étant, les deux équations donnent, en remplaçant ζ par z ,

$$\begin{aligned} a\xi + (h - q\mathfrak{G})\eta &= ax + hy + (g - q\mathfrak{G})z, \\ (h + q\mathfrak{F})\xi + b\eta &= hx + by + (f + q\mathfrak{F})z, \end{aligned}$$

et de là, en remarquant que

$$ab - (h - q\mathfrak{G})(h + q\mathfrak{F}) = \mathfrak{C} + q^2\mathfrak{G}\mathfrak{F} = \mathfrak{C}(1 + q^2\mathfrak{C}),$$

on obtient très-facilement, éliminant successivement η et ξ ,

$$\begin{aligned}(1 + q^2\mathfrak{C})\xi &= (1 + q\mathfrak{H})x + \phi\beta y + (qf + q^2\mathfrak{G})z, \\ (1 + q^2\mathfrak{C})\eta &= -qax + (1 - q\mathfrak{H})y + (-qg + q^2\mathfrak{F})z,\end{aligned}$$

et ensuite

$$\begin{aligned}(1 + q^2\mathfrak{C})x' &= (1 + 2q\mathfrak{H} - q^2\mathfrak{C})x + 2qby + 2(qf + q^2\mathfrak{G})z, \\ (1 + q^2\mathfrak{C})y' &= -2qax + (1 - 2q\mathfrak{H} - q^2\mathfrak{C})y + 2(-qg + q^2\mathfrak{F})z,\end{aligned}$$

c'est-à-dire, en écrivant $z = z' = 1$, les valeurs

$$\begin{aligned}(1 + q^2\mathfrak{C})x' &= (1 + 2q\mathfrak{H} - q^2\mathfrak{C})x + 2qby + 2(qf + q^2\mathfrak{G}) \\ (1 + q^2\mathfrak{C})y' &= -2qax + (1 - 2q\mathfrak{H} - q^2\mathfrak{C})y + 2(-qg + q^2\mathfrak{F})\end{aligned}\quad (4),$$

satisfont identiquement à l'équation

$$(a, b, c, f, g, h \mid x', y', 1)^2 = (a, b, c, f, g, h \mid x, y, 1)^2. \quad (5)$$

En prenant $h = 0$, on a

$$\mathfrak{C} = ab, \quad \mathfrak{G} = -af, \quad \mathfrak{F} = -bg,$$

et les formules deviennent

$$\begin{aligned}(1 + q^2ab)x' &= (1 - q^2ab)x + 2qby + 2q(f - qaf) \\ (1 + q^2ab)y' &= -2qax + (1 - q^2ab)y - 2q(g + qgf)\end{aligned}\quad (6)$$

valeurs qui satisfont identiquement à l'équation

$$(ax^2 + 2gx + l) + (by^2 + 2fy + m) = (ax'^2 + 2gx' + l) + (by'^2 + 2fy' + m) \quad (7),$$

[où pour c j'ai mis $l + m$] et en y écrivant

$$\frac{1 - q^2ab}{1 + q^2ab} = s = \sqrt{1 - abr^2},$$

on obtient des formules qui correspondent précisément aux équations données par Euler pour x, y en termes de a, b .

Londres, 10 Mars, 1857.

197.

NOTE ON THE THEORY OF LOGARITHMS.

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 275–280.]

AN imaginary quantity $x + yi$ may always be expressed in the form

$$x + yi = r(\cos \theta + i \sin \theta) = re^{i\theta},$$

where r is positive, and θ is included between the limits $-\pi$ and $+\pi$. We have in fact

$$r = +\sqrt{x^2 + y^2};$$

and when x is positive,

$$\theta = \tan^{-1} \frac{y}{x};$$

but when x is negative,

$$\theta = \tan^{-1} \frac{y}{x} \pm \pi;$$

where $\tan^{-1}$ denotes an arc between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$, and where the upper or under sign is to be employed according as y is positive or negative. I use for convenience the mark $\equiv$ to denote identity of sign; we may then write

$$\theta = \tan^{-1} \frac{y}{x} + e\pi,$$

where

$$x = +, e = 0, \quad x = -, e = \pm 1 \equiv y.$$

It should be remarked that θ has a unique value except in the single case $x = -, y = 0$, where θ is indeterminately $\pm\pi$. We have, in fact, $\theta = +\pi$ or $\theta = -\pi$ according as x is considered as the limit of $x + yi$, $y = +$, or of $x + yi$, $y = -$. It is natural to write

$$\log(x + yi) = \log r + \theta i,$$

or what is the same thing,

$$\log(x + yi) = \log \sqrt{x^2 + y^2} + i \left(\tan^{-1} \frac{y}{x} + e\pi \right);$$

and I take this equation as the definition of the logarithm of an imaginary quantity. The question then arises, to find the value of the expression

$$\log(x + yi) + \log(x' + y'i) - \log[(x + yi)(x' + y'i)].$$

The preceding definition is, in fact, in the case of x positive, that given by M. Cauchy in the *Exercices de Mathématique*, vol. I. [1826]; and he has there shown that $x, x', xx' - yy'$ being all of them positive, the above-mentioned expression reduces itself to zero. The general definition is that given in my *Mémoire sur quelques Formules du Calcul Intégral*, Liouville, vol. XII. [1847], p. 231 [49]; but I was wrong in asserting that the expression always reduced itself to zero. We have, in fact, in general

$$\tan^{-1} \alpha + \tan^{-1} \beta = \tan^{-1} \frac{\alpha + \beta}{1 - \alpha\beta},$$

when $1 - \alpha\beta$ is positive; but when $1 - \alpha\beta$ is negative (which implies that α, β have the same sign), then

$$\tan^{-1} \alpha + \tan^{-1} \beta = \tan^{-1} \frac{\alpha + \beta}{1 - \alpha\beta} \pm \pi,$$

where the upper or under sign is to be employed according as α and β are positive or negative; or what is the same thing,

$$\tan^{-1} \alpha + \tan^{-1} \beta = \tan^{-1} \frac{\alpha + \beta}{1 - \alpha\beta} + e''\pi,$$

where

$$1 - \alpha\beta = +, \quad e'' = 0; \quad 1 - \alpha\beta = -, \quad e'' = \pm 1 \equiv \alpha \equiv \beta.$$

This being premised, then writing

$$\begin{aligned} \log(x + yi) &= \log \sqrt{x^2 + y^2} + \left(\tan^{-1} \frac{y}{x} + e\pi \right) i, \\ \log(x' + y'i) &= \log \sqrt{x'^2 + y'^2} + \left(\tan^{-1} \frac{y'}{x'} + e'\pi \right) i, \\ \log(x + yi)(x' + y'i) &= \log[(xx' - yy') + (xy' + yx')i] \\ &= \log \sqrt{x^2 + y^2} \sqrt{x'^2 + y'^2} + \left(\tan^{-1} \frac{xy' + x'y}{xx' - yy'} + e''\pi \right) i, \end{aligned}$$

we find

$$\log(x + yi) + \log(x' + y'i) - \log(x + yi)(x' + y'i) = (e + e' - e'' + e''')\pi i.$$

Hence, considering the different cases:

I. $x = +, x' = +, xx' - yy' = +,$

$$e = 0, \quad e' = 0, \quad e'' = 0, \quad e''' = 0,$$

and therefore $e + e' - e'' + e''' = 0$.

II. $x = +, x' = +, xx' - yy' = -,$

$$e = 0, \quad e' = 0, \quad e'' = \pm 1 \equiv (xy' + x'y) \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right), \quad e''' = \pm 1 \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right),$$

and therefore $e + e' - e'' + e''' = 0$.

III. $x = +, x' = -, xx' - yy' = +,$

$$e = 0, \quad e' = \pm 1 \equiv y' \equiv -\frac{y'}{x'} \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right), \quad e'' = 0, \quad e''' = \pm 1 \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right),$$

and therefore $e + e' - e'' + e''' = 0$.

IV. $x = +, x' = -, xx' - yy' = -,$

$$e = 0, \quad e' = \pm 1 \equiv y' \equiv -\frac{y'}{x'},$$

$$e'' = \pm 1 \equiv xy' + x'y \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right), \quad e''' = 0,$$

and therefore

$$e + e' - e'' + e''' = 0 \text{ if } \frac{y'}{x'} \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right),$$

but

$$e + e' - e'' + e''' = \pm 2 \equiv \frac{y'}{x'} \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right) \text{ if } \frac{y'}{x'} \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

V. $x = -, x' = +, xx' - yy' = +,$

$$e = \pm 1 \equiv y \equiv -\frac{y}{x} \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right), \quad e' = 0, \quad e'' = 0, \quad e''' = \pm 1 \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right),$$

and therefore $e + e' - e'' + e''' = 0$.

VI. $x = -, x' = +, xx' - yy' = -,$

$$e = \pm 1 \equiv y \equiv -\frac{y}{x}, \quad e' = 0,$$

$$e'' = \pm 1 \equiv (xy' + x'y) \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right), \quad e''' = 0,$$

and therefore

$$e + e' - e'' + e''' = 0 \text{ if } \frac{y}{x} \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right),$$

but

$$e + e' - e'' + e''' = \pm 2 \equiv \frac{y}{x} \equiv -\frac{y'}{x'} \text{ if } \frac{y}{x} \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

VII. $x = -, x' = -, xx' - yy' = +,$

$$e = \pm 1 \equiv y \equiv -\frac{y}{x}, \quad e' = \pm 1 \equiv y' \equiv -\frac{y'}{x'}, \quad e'' = 0, \quad e''' = 0;$$

and therefore

$$e + e' - e'' + e''' = 0 \text{ if } \frac{y}{x} \equiv -\frac{y'}{x'},$$

but

$$e + e' - e'' + e''' = \pm 2 \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right) \text{ if } \frac{y}{x} \equiv \frac{y'}{x'}.$$

VIII. $x = -, x' = -, xx' - yy' = -,$

$$e = \pm 1 \equiv y \equiv -\frac{y}{x}, \quad e' = \pm 1 \equiv y' \equiv -\frac{y'}{x'},$$

$$e'' = \pm 1 \equiv (xy' + x'y) \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right), \quad e''' = \pm 1 \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right),$$

and therefore

$$e + e' - e'' + e''' = \pm 2 \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

Hence writing

$$\log(x + yi) + \log(x' + y'i) - \log(x + yi)(x' + y'i) = E\pi i,$$

we have $E = 0$, except in the following cases, viz.

1. (See IV.)

$$x = +, x' = -, xx' - yy' = -, \frac{y'}{x'} \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right),$$

where

$$E = \pm 2 \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

2. (See VI.)

$$x = -, x' = +, xx' - yy' = -, \frac{y}{x} \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right),$$

where

$$E = \pm 2 \equiv \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

3. (See VII.)

$$x = -, x' = -, xx' - yy' = +, \frac{y}{x} \equiv \frac{y'}{x'},$$

where

$$E = \pm 2 \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

4. (See VIII.)

$$x = -, x' = -, xx' - yy' = -,$$

where

$$E = \pm 2 \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

It thus appears that when the real parts x , x' , $xx' - yy'$ are all three of them positive, or any two of them positive and the third negative, E is equal to zero, or the logarithm of the product is equal to the sum of the logarithms of the factors; but that if the real parts are one of them positive and the other two of them negative, then if a certain relation between the real and imaginary parts is satisfied, but not otherwise, the property holds; and if the real parts are all three of them negative, the property does not hold in any case.

The preceding results do not apply to the case where any one of the arguments $x + yi$, $x' + y'i$, $(x + yi)(x' + y'i)$ is real and negative, for no definition applicable to such case has been given of a logarithm. If, however, we assume as a definition that the logarithm of a negative real quantity is equal to the logarithm of the corresponding positive quantity, then in the case $x = -$, $y = 0$, we have

$$\log x + \log(x' + y'i) - \log x(x' + y'i) = e\pi i, \quad e = \pm 1 \equiv y';$$

an equation which is, in fact, equivalent to

$$\log(x' + y'i) - \log[-(x' + y'i)] = e\pi i, \quad e = \pm 1 \equiv y';$$

and in the case $xy' + x'y = 0$, $xx' - yy' = -$, which implies $y \equiv y'$, then

$$\log(x + yi) + \log(x' + y'i) - \log(x + yi)(x' + y'i) = e\pi i, \quad e = \pm 1 \equiv y \text{ or } y';$$

an equation which is in fact equivalent to

$$\log(x + yi) + \log(-x - yi) - \log(-x^2 - y^2) = e\pi i, \quad e = \pm 1 \equiv y.$$

The case where both of the arguments $x + yi$, $x' + y'i$ are real and negative, i.e. $x = -$, $y = 0$, $x' = -$, $y' = 0$, gives of course $\log x + \log x' - \log xx' = 0$, the logarithms of the negative real quantities x , x' being by the definition the same as the logarithms of the corresponding positive quantities. It should, however, be remarked that the definition, ($x = -$), $\log x = \log(-x)$ not only gives for $\log x$ a different value from that which would be obtained from the general definition of a logarithm, by considering $\log x$ as the limit of $\log(x + yi)$, $y = +$, or of $\log(x + yi)$, $y = -$, but gives also a value, which, for the particular case in question, contradicts the fundamental equation $e^{\log x} = x$. It is therefore, I think, better not to establish any definition for the logarithm of a negative real quantity x , but to say that such logarithm is absolutely indeterminate and indeterminable, except in the case where, from the nature of the question, x is considered as the limit of $x + yi$, y positive, or of $x + yi$, y negative.

2, *Stone Buildings*, March 15, 1856.

198.

NOTE UPON A RESULT OF ELIMINATION.

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 378–379.]

If the quadratic function

$$(a, b, c, f, g, h \mid x, y, z)^2$$

break up into factors, then representing one of these factors by $\xi x + \eta y + \zeta z$, and taking any arbitrary quantities α, β, γ , the factor in question, and therefore the quadratic function is reduced to zero by substituting $\beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi$ in the place of x, y, z . Write

$$(a, b, c, f, g, h \mid \beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi)^2 = (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h} \mid \alpha, \beta, \gamma)^2;$$

the coefficients of the function on the right hand are

$$\begin{aligned}\mathfrak{a} &= \cdot \cdot \cdot c\eta^2 + b\zeta^2 - 2f\eta\zeta \cdot \cdot \\ \mathfrak{b} &= c\xi^2 + \cdot \cdot + a\zeta^2 \cdot \cdot - 2g\zeta\xi \cdot \\ \mathfrak{c} &= b\xi^2 + a\eta^2 + \cdot \cdot \cdot \cdot - 2h\xi\eta \\ \mathfrak{f} &= -f\xi^2 \cdot \cdot \cdot - a\eta\zeta + h\zeta\xi + g\xi\eta \\ \mathfrak{g} &= \cdot - g\eta^2 \cdot + h\eta\zeta - b\zeta\xi + f\xi\eta \\ \mathfrak{h} &= \cdot \cdot - h\zeta^2 + g\eta\zeta + f\zeta\xi - c\xi\eta;\end{aligned}$$

and it is to be remarked that we have identically

$$\begin{aligned}\mathfrak{a}\xi + \mathfrak{h}\eta + \mathfrak{g}\zeta &= 0, \\ \mathfrak{h}\xi + \mathfrak{b}\eta + \mathfrak{f}\zeta &= 0, \\ \mathfrak{g}\xi + \mathfrak{f}\eta + \mathfrak{c}\zeta &= 0.\end{aligned}$$

Hence of the six equations, $\mathfrak{a} = 0, \mathfrak{b} = 0, \mathfrak{c} = 0, \mathfrak{f} = 0, \mathfrak{g} = 0, \mathfrak{h} = 0$, any three (except $\mathfrak{a} = 0, \mathfrak{h} = 0, \mathfrak{g} = 0$, or $\mathfrak{h} = 0, \mathfrak{b} = 0, \mathfrak{f} = 0$, or $\mathfrak{g} = 0, \mathfrak{f} = 0, \mathfrak{c} = 0$) imply the remaining three.

If from the six equations we eliminate ξ^2 , η^2 , &c., we obtain

$$\square = \begin{vmatrix} \cdot & c & b & -2f & \cdot & \cdot \\ c & \cdot & a & \cdot & -2g & \cdot \\ b & a & \cdot & \cdot & \cdot & -2h \\ -f & \cdot & \cdot & -a & h & g \\ \cdot & -g & \cdot & h & -b & f \\ \cdot & \cdot & -h & g & f & -c \end{vmatrix} = 0;$$

and the equation $\square = 0$ is therefore the result of the elimination of ξ , η , ζ from any three (other than the excepted combinations) of the six equations. But from what precedes, it appears that the equation $\square = 0$ must be satisfied when the quadratic function breaks up into factors, and consequently $\square$ must contain as a factor the discriminant

$$K = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

of the quadratic function. This agrees perfectly with the results obtained long ago by Prof. Sylvester in his paper, “Examples of the Dialectic Method of Elimination as applied to Ternary Systems of Equations,” *Camb. Math. Journ.* vol. II. p. 232; but according to the assumption there made, the value of $\square$ would be (to a numerical factor près) $= abcK$. The correct value is found by actual development shown to be $\square = -2K^2$. It would be interesting to show *à priori* that $\square$ contains K^2 as a factor.

2, *Stone Buildings*, March 28, 1856.

199.

NOTE ON THE THEORY OF ELLIPTIC MOTION.

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 425–428.]

IF, as usual, r , θ denote the radius vector and longitude, and μ the central mass, then the Vis Viva and Force function are respectively

$$T = \frac{1}{2}(r'^2 + r^2\theta'^2), \quad U = \frac{\mu}{r};$$

and writing

$$\frac{dT}{dr'} = r' = p, \quad \frac{dT}{d\theta'} = r^2\theta' = q,$$

we have $r' = p$, $\theta' = \frac{q}{r^2}$, and $T = \frac{1}{2}\left(p^2 + \frac{q^2}{r^2}\right)$, whence, putting $H = T - U$, the value of H is

$$H = \frac{1}{2}\left(p^2 + \frac{q^2}{r^2}\right) - \frac{\mu}{r};$$

and by Sir W. R. Hamilton's theory, the equations of motion are

$$\frac{dr}{dt} = \frac{dH}{dp}, \quad \frac{dp}{dt} = -\frac{dH}{dr},$$

$$\frac{d\theta}{dt} = \frac{dH}{dq}, \quad \frac{dq}{dt} = -\frac{dH}{d\theta};$$

or substituting for H its value, the equations of motion are

$$\begin{aligned}\frac{dr}{dt} &= p, & \frac{d\theta}{dt} &= \frac{q}{r^2}, \\ \frac{dp}{dt} &= \frac{q^2}{r^3} - \frac{\mu}{r^2}, & \frac{dq}{dt} &= 0.\end{aligned}$$

Putting, as usual, $\mu = n^2 a^3$, and introducing the eccentric anomaly u , which is given as a function of t by means of the equation

$$nt + c - \varpi = u - e \sin u,$$

(so that $\frac{du}{dt} = \frac{n}{1 - e \cos u}$), the integral equations are

$$\begin{aligned}q &= na^2 \sqrt{1 - e^2}, \\ p &= \frac{nae \sin u}{1 - e \cos u}, \\ r &= a(1 - e \cos u), \\ \theta - \varpi &= \tan^{-1} \left(\frac{\sqrt{1 - e^2} \sin u}{\cos u - e} \right),\end{aligned}$$

where the constants of integration a, e, c, ϖ denote as usual the mean distance, the eccentricity, the mean anomaly at epoch, and the longitude of pericentre.

Suppose that $q_0, p_0, r_0, \theta_0, u_0$ correspond to the time t_0 (q is constant, so that $q_0 = q$), and write

$$V = na^2(u - u_0 + e \sin u - e \sin u_0);$$

joining to this the equations

$$\begin{aligned}r &= a(1 - e \cos u), & r_0 &= a(1 - e \cos u_0), \\ \theta - \theta_0 &= \tan^{-1} \left(\frac{\sqrt{1 - e^2} \sin u}{\cos u - e} \right) - \tan^{-1} \left(\frac{\sqrt{1 - e^2} \sin u_0}{\cos u_0 - e} \right),\end{aligned}$$

u, u_0, e will be functions of $a, r, r_0, \theta, \theta_0$, and consequently (n being throughout considered as a function of a) V will be a function of $a, r, r_0, \theta, \theta_0$. The function V so expressed as a function of $a, r, r_0, \theta, \theta_0$ is, in fact, the characteristic function of Sir W. R. Hamilton, and according to his theory we ought to have

$$dV = \frac{3}{2}n^2 a(t - t_0) da + p dr + q d\theta - p_0 dr_0 - q d\theta_0.$$

To verify this, I form the equation

$$\begin{aligned}
 dV = & \frac{3}{2}na(u - u_0 + e \sin u - e \sin u_0) da \\
 & + na^2[(1 - e \cos u) du - (1 - e \cos u_0) du_0] \\
 & + na^2(\sin u - \sin u_0) de \\
 & + \frac{nae \sin u}{1 - e \cos u} [dr - (1 - e \cos u) da - ae \sin u du + a \cos u de] \\
 & - \frac{nae \sin u_0}{1 - e \cos u_0} [dr_0 - (1 - e \cos u_0) da - ae \sin u_0 du_0 + a \cos u_0 de] \\
 & + na^2 \sqrt{1 - e^2} \left\{ d\theta - \frac{1}{\sqrt{1 - e^2}(1 - e \cos u)} [(1 - e^2) du + \sin u a de] \right\} \\
 & - na^2 \sqrt{1 - e^2} \left\{ d\theta_0 - \frac{1}{\sqrt{1 - e^2}(1 - e \cos u_0)} [(1 - e^2) du_0 + \sin u_0 a de] \right\};
 \end{aligned}$$

the coefficient of du on the right-hand side is

$$\begin{aligned}
 & na^2(1 + e \cos u) - \frac{na^2 e^2 \sin^2 u}{1 - e \cos u} - \frac{na^2(1 - e^2)}{1 - e \cos u} \\
 & = na^2 \left(1 + e \cos u - \frac{1 - e^2 + e^2 \sin^2 u}{1 - e \cos u} \right), \\
 & = na^2 \left(1 + e \cos u - \frac{1 - e^2 \cos^2 u}{1 - e \cos u} \right),
 \end{aligned}$$

which vanishes, and similarly the coefficient of du_0 also vanishes; the coefficient of de is the difference of two parts, the first of which is

$$\begin{aligned}
 & na^2 \sin u + \frac{na^2 e \sin u \cos u}{1 - e \cos u} - \frac{na^2 \sin u}{1 - e \cos u} \\
 & = na^2 \sin u \left(1 - \frac{1 - e \cos u}{1 - e \cos u} \right),
 \end{aligned}$$

which vanishes, and the second part in like manner also vanishes; the coefficient of da is the difference of two parts, the first of which is

$$\frac{3}{2}na(u - e \sin u) - nae \sin u \cdot \frac{1}{2}na(u - e \sin u),$$

and the second is the like function of u_0 ; the entire coefficient therefore is

$$\frac{3}{2}na(u - u_0 + e \sin u - e \sin u_0).$$

We have therefore

$$\begin{aligned}
 dV = & \frac{3}{2}na(u - u_0 + e \sin u - e \sin u_0) da \\
 & + \frac{nae \sin u}{1 - e \cos u} dr - na^2 \sqrt{1 - e^2} d\theta \\
 & - \frac{nae \sin u_0}{1 - e \cos u_0} dr_0 - na^2 \sqrt{1 - e^2} d\theta_0,
 \end{aligned}$$

or what is the same thing,

$$dV = \frac{3}{2}n^2a(t - t_0) da + p dr + q d\theta - p_0 dr_0 - q d\theta_0,$$

the equation which was to be verified.

2, *Stone Buildings, March 28, 1856.*

200.

ON THE CONES WHICH PASS THROUGH A GIVEN CURVE OF THE THIRD ORDER IN SPACE.

[From the *Philosophical Magazine*, vol. XII. (1856), pp. 20–22.]

THE following investigation is connected with the theory of the cubic

$$(a, b, c, d \mid x, y)^3,$$

and in particular with a theorem that the determinant formed with the second differential coefficients of the discriminant gives the square of the discriminant. Consider the coefficients a, b, c, d as linear functions of coordinates, the equations

$$ac - b^2 = 0, \quad bd - c^2 = 0, \quad ad - bc = 0$$

(equivalent, of course, to two equations) belong to a curve of the third order in space, the edge of regression of the developable surface obtained by putting the discriminant equal to zero, or which has for its equation

$$-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2 = 0;$$

and, moreover, the above forms are the general representations of any curve of the third order, and developable surface of the fourth order. The question arises, to find the equation of the cone of the third order having an arbitrary point for its vertex, and passing through the curve of the third order. This may be done by Joachimsthal's method: let $\alpha, \beta, \gamma, \delta$ be the values of a, b, c, d at the vertex of the cone, and in the equations of the curve, say in

$$ac - b^2 = 0, \quad bd - c^2 = 0,$$

for a, b, c, d write $ua + v\alpha, ub + v\beta, uc + v\gamma, ud + v\delta$; if from the equations so obtained u, v are eliminated, the resulting equation will be that of the cone of the third order.

The substitutions in question give

$$\begin{aligned} Lu^2 + 2Muv + Nv^2 &= 0, \\ L'u^2 + 2M'uv + N'v^2 &= 0, \end{aligned}$$

where

$$\begin{aligned} L &= 2(ac - b^2), & M &= a\gamma + c\alpha - 2b\beta, & N &= 2(\alpha\gamma - \beta^2), \\ L' &= 2(bd - c^2), & M' &= b\delta + d\beta - 2c\gamma, & N' &= 2(\beta\delta - \gamma^2); \end{aligned}$$

the result of the elimination is

$$4(LN' - M^2)(L'N - M'^2) - (LN' + L'N - 2MM')^2 = 0,$$

and we have

$$\begin{aligned} LN - M^2 &= 4(b\gamma - c\beta)(a\beta - b\alpha) - (c\alpha - a\gamma)^2, \\ L'N' - M'^2 &= 4(c\delta - d\gamma)(b\gamma - c\beta) - (d\beta - b\delta)^2, \\ LN' + L'N - 2MM' &= 4(b\gamma - c\beta)^2 + 4(a\beta - b\alpha)(c\delta - d\gamma) + 2(c\alpha - a\gamma)(d\beta - b\delta). \end{aligned}$$

Write for shortness

$$\begin{aligned} b\gamma - c\beta &= l, & c\alpha - a\gamma &= m, & a\beta - b\alpha &= n, \\ a\delta - d\alpha &= f, & b\delta - d\beta &= g, & c\delta - d\gamma &= h, \end{aligned}$$

values which give $lf + mg + nh = 0$. Then forming the expression

$$4(4ln - m^2)(4lh - g^2) - (4hl + lm + 4ln + mg + 2nh)^2,$$

this is equal to ⁽¹⁾

$$-16l(l^2h - nfh + lmg - 2lnh + ng^2 + m^2h),$$

and the equation of the cone is

$$l^2h - nfh + lmg - 2lnh + ng^2 + m^2h = 0;$$

or, substituting and expanding,

$$\begin{aligned} &(a\delta - \beta)[\dots] - \\ &\quad (\text{long expansion in } a, b, c, d, \alpha, \beta, \gamma, \delta) \\ &(b^2d - bc^2)\delta^3 - 3(b\delta - c\gamma)^2a^2c + \dots = 0. \end{aligned}$$

[Figure: long expansion preserved as in scan; the printed form spans many lines and matches the symbolic identity reduced below.]

¹ With respect to the occurrence of the factor $l(= b\gamma - c\beta)$, it is worth noticing, that, putting $b = k\beta$, $c = k\gamma$, we have identically

$$\begin{aligned} Lu^2 + 2Muv + Nv^2 &= 2[(\alpha\gamma - b\beta)u + (\alpha\gamma - \beta^2)v](ku + v), \\ L'u^2 + 2M'uv + N'v^2 &= 2[(\beta\delta - c\gamma)u + (\beta\delta - \gamma^2)v](ku + v), \end{aligned}$$

i.e. the two functions will contain a common factor if $b = k\beta$, $c = k\gamma$, or what is the same thing, if $b\gamma - c\beta = 0$. But if the functions contain a common factor, their resultant vanishes, i.e. the resultant will vanish in virtue of the relation $b\gamma - c\beta = 0$, or what is the same thing, $b\gamma - c\beta$ is a factor of the resultant.

Now putting

$$bd - c^2 = p, \quad bc - ad = q, \quad ac - b^2 = r,$$

and in like manner

$$\beta\delta - \gamma^2 = P, \quad \beta\gamma - \alpha\delta = Q, \quad \alpha\gamma - \beta^2 = R,$$

and reducing, the final result may be expressed in the form

$$\begin{aligned} &P \{ -2\delta bp + (\gamma a - 2\beta b)q + (\delta a - \gamma b)r \} \\ &\quad + Q \{ -\alpha cp + (\gamma b - \beta c)q + \alpha br \} \\ &+ R \{ -(\alpha d + \beta c)p + (\beta d - 2\gamma c)q + 2\delta cr \} = 0, \end{aligned}$$

where a, b, c, d are current coordinates, and p, q, r are quadratic functions of a, b, c, d . The equation is (as it should be) satisfied by the equations ($p = 0, q = 0, r = 0$) of the given curve; it is also satisfied *per se* when $P = 0, Q = 0, R = 0$, i.e. when the vertex is a point on the curve; this indicates a change in form of the equation, and in fact the cone is in this case of the second order only. Suppose that the coordinates of the vertex are in this case given by $\frac{a}{\beta} = \frac{b}{\gamma} = \frac{c}{\delta} = \frac{1}{\sigma}$ (σ an arbitrary quantity), it may be easily shown that the equation of the cone is

$$p + \sigma q + \sigma^2 r = 0;$$

or at full length

$$(bd - c^2) + \sigma(bc - ad) + \sigma^2(ac - b^2) = 0;$$

in fact this equation is evidently that of a surface of the second order passing through the curve; and there is no difficulty in showing that it is a cone.

2, *Stone Buildings, May 1, 1856.*

201.

SECOND NOTE ON THE THEORY OF LOGARITHMS.

[From the *Philosophical Magazine*, vol. XII. (1856), pp. 354–360.]

THE theory of logarithms, as developed in my first note (*Phil. Mag.* April 1856), [197] may be exhibited in a clearer light by considering, instead of $\log a + \log b - \log ab = E\pi i$, the equation $\log \frac{b}{a} = \log b - \log a + E\pi i$, a form which more readily enables the accounting *à priori* for the discontinuity in the value of E . Writing then for b, a the complex values $x' + y'i, x + yi$, we have

$$\log \frac{x' + y'i}{x + yi} = \log(x' + y'i) - \log(x + yi) + E\pi i;$$

where, according to the assumed definition of a logarithm,

$$\log(x + yi) = \log \sqrt{x^2 + y^2} + i \left(\tan^{-1} \frac{y}{x} + e\pi \right),$$

in which $\log \sqrt{x^2 + y^2}$ is the real logarithm of $\sqrt{x^2 + y^2}$, and $\tan^{-1} \frac{y}{x}$ is an arc between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$. The coefficient e is equal to zero when x is positive; but when x is negative, then $e = +1$ or -1 , according as y is positive or negative, i.e. we have

$$x = +, e = 0,$$

$$x = -, e = \pm 1 \equiv y;$$

and of course the other logarithms in the equation have an analogous signification.

Hence, attending to the equation

$$\tan^{-1} \alpha - \tan^{-1} \beta = \tan^{-1} \frac{\beta - \alpha}{1 + \alpha\beta} + e''\pi,$$

where, when $1 + \alpha\beta$ is positive, e'' is equal to zero; but when $1 + \alpha\beta$ is negative, e'' is equal to $+1$ or -1 , according as $\beta - \alpha$ is positive or negative; or what is the same thing (α, β being of opposite signs when $1 + \alpha\beta$ is negative),

$$1 + \alpha\beta = +, \quad e'' = 0,$$

$$1 + \alpha\beta = -, \quad e'' = \pm 1 \equiv \beta - \alpha \equiv -\alpha \equiv \beta,$$

we find

$$E = e - e' + e'' - e''',$$

where e, e', e'', e''' are defined by the conditions

$$\begin{aligned} x = +, \quad e &= 0, \\ x = -, \quad e &= \pm 1 \equiv y, \\ x' = +, \quad e' &= 0, \\ x' = -, \quad e' &= \pm 1 \equiv y', \\ xx' + yy' = +, \quad e'' &= 0, \\ xx' + yy' = -, \quad e'' &= \pm 1 \equiv xy' - x'y, \\ 1 + \frac{y}{x} \frac{y'}{x'} = +, \quad e''' &= 0, \\ 1 + \frac{y}{x} \frac{y'}{x'} = -, \quad e''' &= \pm 1 \equiv \frac{y'}{x'} - \frac{y}{x} \equiv -\frac{y}{x} \equiv \frac{y'}{x'}. \end{aligned}$$

Suppose, to fix the ideas, that x, y are each of them positive, we have $e = 0$; and considering the several cases:

1. $x' = +, y' = +$.

Here $xx' + yy' = +$, $1 + \frac{y}{x} \frac{y'}{x'} = +$; and consequently not only $e' = 0$, but also $e'' = 0$, $e''' = 0$, and thence $E = 0$.

2. $x' = +, y' = -$.

Here $e' = 0$. Moreover, xx' being positive, $xx' + yy'$ and $1 + \frac{y}{x} \frac{y'}{x'}$ will have the same sign. If they are both positive, then $e'' = 0$, $e''' = 0$; but if they are both negative, then

$$e'' = \pm 1 \equiv xy' - x'y \equiv \left(\frac{y'}{x'} - \frac{y}{x} \right)$$

(since $xx' = +$) and $e''' = \pm 1 \equiv \left(\frac{y'}{x'} - \frac{y}{x} \right)$, i.e. $e'' = e'''$. Hence in either case we have $E = 0$.

3. $x' = -, y' = +$.

Here $e' = \pm 1 \equiv y'$, $\frac{y'}{x'} = -\frac{y'}{x}$, i.e. $e' = 1$. Also xx' being negative, $xx' + yy'$ and $1 + \frac{y}{x} \frac{y'}{x'}$ will have opposite signs. Suppose first $xx' + yy'$ is positive, then $e'' = 0$. And $1 + \frac{y}{x} \frac{y'}{x'}$ being negative, we have

$$e''' = \pm 1 \equiv \frac{y'}{x'}, \quad e''' = 1 \equiv -\frac{y'}{x},$$

i.e. $e''' = 1$. But if $xx' + yy'$ is negative, then $e'' = \pm 1 \equiv xy' - x'y \equiv \left(\frac{y'}{x'} - \frac{y}{x}\right)$ (since xx' is negative) $= +$,

i.e. $e'' = +1$. And $1 + \frac{y}{x} \frac{y'}{x'}$ being positive, we have $e''' = 0$. Hence in each case $E = 0$.

4. $x' = -, y' = -$.

Here $e' = \pm 1 \equiv y' = -$, i.e. $e' = -1$. Also xx' and yy' being each negative, $xx' + yy'$ will be negative, and therefore

$$e'' = \pm 1 \equiv xy' - x'y = -\left(\frac{y'}{x'} - \frac{y}{x}\right),$$

i.e. if $\frac{y'}{x'} > \frac{y}{x}$, then $e'' = -1$; but if $\frac{y'}{x'} < \frac{y}{x}$, then $e'' = +1$. And $1 + \frac{y}{x} \frac{y'}{x'}$ being positive, $e''' = 0$. Hence if $\frac{y'}{x'} > \frac{y}{x}$, then $E = 1 - 1 = 0$; but if $\frac{y'}{x'} < \frac{y}{x}$, then $E = 1 + 1 = 2$.

Consider (x, y) , (x', y') as the rectangular coordinates of two points A , A' . In the case which has been considered, the point A has been taken in the positive quadrant; and the preceding discussion shows that we have always $E = 0$, except in the case where the finite line AA' meets the negative portion of the axis of x , in which case we have $E = +2$. The same thing is true generally in whichever quadrant

[Figure: rectangular coordinate diagram with two points $A = (x, y)$ in the upper-right quadrant and $A' = (x', y')$ in the lower-left quadrant; the line AA' crosses the negative x -axis between them.]

A is situated, i.e. we have always $E = 0$, except in the cases in which the finite line AA' meets the negative portion of the axis of x . But when this happens, then if the line AA' , considered as drawn from A to A' , passes from above to below the axis of x , we have $E = +2$; but if the line AA' , considered as drawn from A to A' , passes from below to above the axis of x , then $E = -2$. So that treating the points

A, A' as the geometrical representations of the complex numbers $x+yi, x'+y'i$, we have in an exceedingly simple form the precise determination of the discontinuous number E ($= 0$ or ± 2) in the formula

$$\log \frac{x' + y'i}{x + yi} = \log(x' + y'i) - \log(x + yi) + E\pi i.$$

Consider in general the definite integral

$$\int_z^{z'} \phi u \, du,$$

where z, z' are complex numbers of the form $x+yi, x'+y'i$; and take A, A' as the geometrical representations of these limits, and the variable point P as the geometrical representation of the complex variable u . The value of the definite integral will depend to a certain extent on the series of values which we suppose u successively to assume in passing from z to z' , or what is the same thing, on the path of the variable point P from A to A' . For (excluding altogether the case in which the path passes through a point for which ϕu becomes infinite) it is well known that the value of the definite integral is the same for any two paths which do not include between them a point for which ϕu becomes infinite; but when this condition is not satisfied, then the value of the definite integral is not in general the same for the two paths.

[Figure: a path between two points A and A' in the complex plane, with the variable point P moving along the path.]

In order therefore to give a precise signification to the notations, we must fix the path of the point P , and it is natural to assume that the path is a right line (of course there are an infinity of paths which give the same value to the definite integral, or as we may call them, paths equivalent to the right line; but the consideration of these would be a needless complication of the definition, and it is better to attend to the single path the right line). The definition is at once converted into an analytical one; we have only to assume $u = z + r(z' - z)$, and to suppose that the new variable r passes from $r = 0$ (which gives $u = z$) through real

¹ The theorem is, I believe, due to M. Cauchy. See the memoir of M. Puiseux, *Recherches sur les Fonctions Algébriques*, Liouville, vol. xv. [1850] pp. 365–480, where the subject is elaborately discussed.

values to $r = 1$ (which gives $u = z'$), i.e. we have as the equivalent analytical definition of the definite integral between the complex limits z, z' the equation

$$\int_z^{z'} \phi u \, du = (z' - z) \int_0^1 \phi[z + r(z' - z)] \, dr,$$

where the new variable r is real. The only restriction is, that ϕu must not become infinite for any value of u along the path in question, i.e. $\phi[z + r(z' - z)]$ must not become infinite for any real value of r between the limits $r = 0, r = 1$.

Suppose next, the path being defined as above, or in any other manner, that $\phi_1 u$ is a function of u such that $\phi'_1 u = \phi u$. Then if $\phi_1 u$ is continuous along the entire path, we have

$$\int_z^{z'} \phi u \, du = \phi_1 z' - \phi_1 z,$$

but if $\phi_1 u$ is discontinuous at any points of the path, e.g. at the point $u = u_1$ and at no other point, then

$$\int_z^{z'} \phi u \, du = \phi_1 z' - \phi_1(u_1 + \alpha) + \phi_1(u_1 - \alpha) - \phi_1 z,$$

where $u_1 - \alpha, u_1 + \alpha$ are values indefinitely near to u_1 , the path being from z through $u_1 - \alpha$ to $u_1 + \alpha$ and thence to z' . Or if we represent the break $\phi_1(u_1 + \alpha) - \phi_1(u_1 - \alpha)$ by the symbol ∇ , then we have

$$\int_z^{z'} \phi u \, du = \phi_1 z' - \phi_1 z - \nabla.$$

Suppose now

$$\zeta = \tan^{-1} \frac{y}{x} + e\pi,$$

where, as before, $\tan^{-1} \frac{y}{x}$ denotes an arc between the limits $-\frac{1}{2}\pi, +\frac{1}{2}\pi$, and

$$x = +, e = 0,$$

$$x = -, e = \pm 1 \equiv y;$$

and to fix the ideas, consider ζ as the z -coordinate of a surface, the other two coordinates being x and y . If x be negative and y be indefinitely small and positive, then $e = +1$, and we have $\zeta = \pi$; but if (x being still negative) y be indefinitely small and negative, then $e = -1$, and therefore $\zeta = -\pi$, i.e. there is a break or abrupt increment 2π of the coordinate ζ in passing across the negative part of the axis of x from a negative to a positive value of y , or, as we have before called it, from below to above; this is the only discontinuity in the surface, the form of the surface being, in fact, what is intended to be represented in the annexed figure.

Suppose now $z' = x' + y'i, z = x + yi$, and consider the definite integral

$$\int_z^{z'} \frac{du}{u},$$

the path being, as before, a right line. We have, by the equivalent analytical definition,

$$\int_z^{z'} \frac{du}{u} = (z' - z) \int_0^1 \frac{dr}{z + r(z' - z)},$$

where the new variable r is real. And in like manner considering the integral $\int_1^{z'/z} \frac{du}{u}$, the path being in this case also a right line, we have

$$\int_1^{z'/z} \frac{du}{u} = \left(\frac{z'}{z} - 1 \right) \int_0^1 \frac{dr}{1 + r\left(\frac{z'}{z} - 1\right)},$$

where the new variable r is real. The two integrals in r are identically the same, and consequently we have in every case

$$\int_z^{z'} \frac{du}{u} = \int_1^{z'/z} \frac{du}{u}.$$

Now $\log' u = \frac{1}{u}$; and in passing from $u = 1$ to $u = \frac{z'}{z}$, there is no discontinuity in the value of $\log u$,

$$= \log \sqrt{p^2 + q^2} + i \left(\tan^{-1} \frac{q}{p} + e\pi \right),$$

if for the moment $u = p + qi$; hence the value of the integral on the left-hand side is simply $\log \frac{z'}{z}$. The value of the integral on the right-hand side is in like manner $\log z' - \log z$, in the case in which the finite right line from $u = z$ to $u = z'$ does not meet the negative part of the axis of x ; but when this happens, there is a discontinuity in the value of the logarithm, and the integral on the right-hand side

will be $\log z' - \log z + 2\pi i$, or $\log z' - \log z - 2\pi i$, according as the right line considered as drawn from z to z' passes from below to above or from above to below the negative part of the axis of x . We have therefore in every case (E being defined as above) $\log z' - \log z + E\pi i$ for the value of the integral on the right-hand side, and the relation between the two integrals gives, as it ought to do, the equation

$$\log \frac{z'}{z} = \log z' - \log z + E\pi i,$$

or, in the form in which it was before written,

$$\log \frac{x' + y'i}{x + yi} = \log(x' + y'i) - \log(x + yi) + E\pi i.$$

The preceding discussion shows that the discontinuity in the value of E ($= 0$ or ± 2) arises from, or is most intimately connected with, the geometrical discontinuity which necessarily exists in the surface $z = \tan^{-1} \frac{y}{x}$, whenever we define the symbol $\tan^{-1}$ in such manner as to give a unique value to the coordinate z .

2, *Stone Buildings*, Sept. 19, 1856.